

Assignment 6

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Data Exploration

1 Dataset Analysis

This section provides a descriptive overview of the dataset used for the corporate bankruptcy prediction models. The dataset integrates firm-level financial ratios from Compustat and market-based indicators from CRSP, spanning the years 1964–2023.

1.1 Bankruptcies Over Time

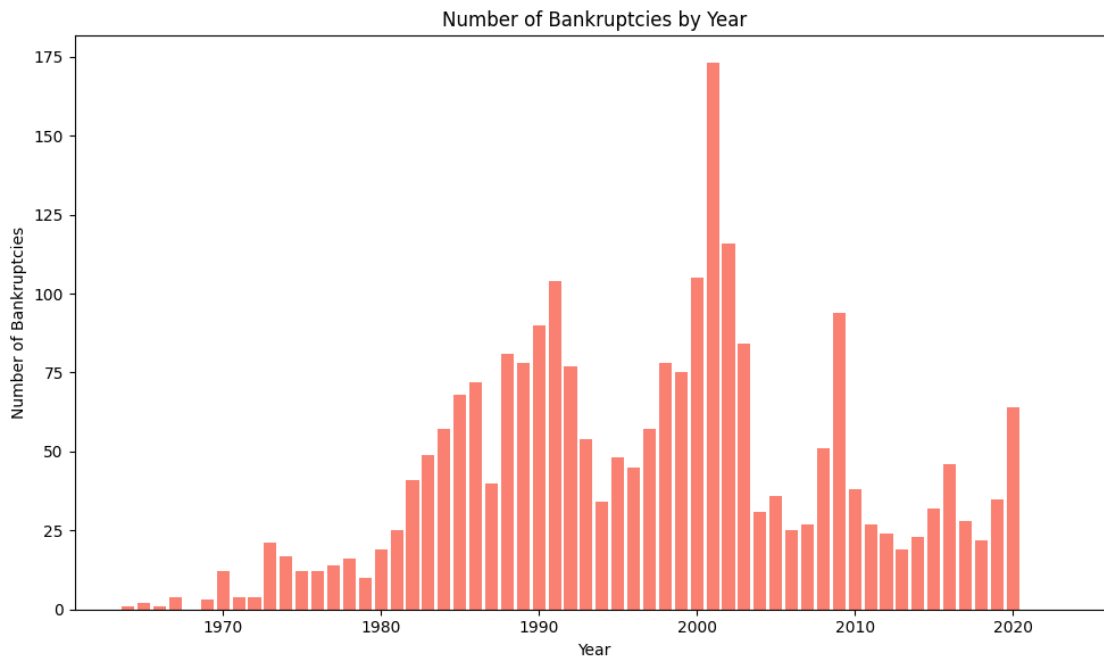


Figure 1: Number of Bankruptcies by Year

The number of recorded bankruptcies varies across years, with relatively few observations in the early part of the sample and a denser concentration after 1990. The increase in the post-1990 period primarily reflects improved data coverage and a larger population of publicly traded firms. This period provides a richer representation of firm characteristics and offers the main basis for model estimation and testing.

1.2 Distribution of Firm-Years

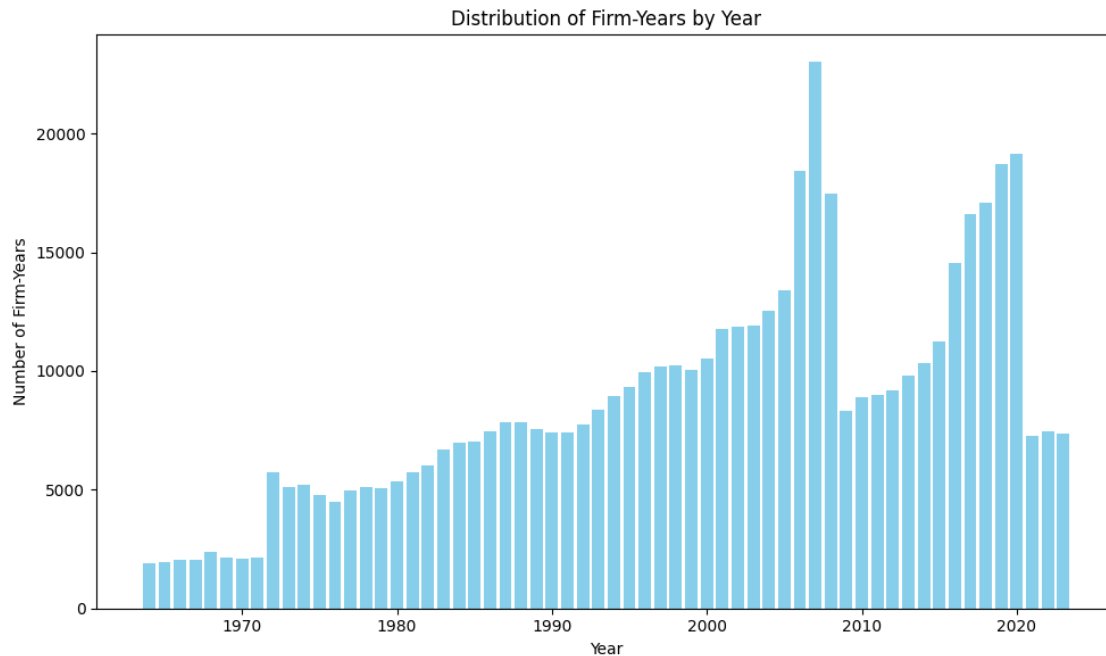


Figure 2: Distribution of Firm-Years by Year

The number of firm-year observations increases markedly after 1990. While earlier decades contain a smaller subset of firms, the post-1990 data include tens of thousands of observations per year. This concentration ensures that the estimation sample is dominated by recent years, allowing models to learn from a more stable and comprehensive financial reporting environment. It also implies that model performance primarily reflects patterns in the modern period of financial disclosure and market structure.

1.3 Correlation Structure of Predictors

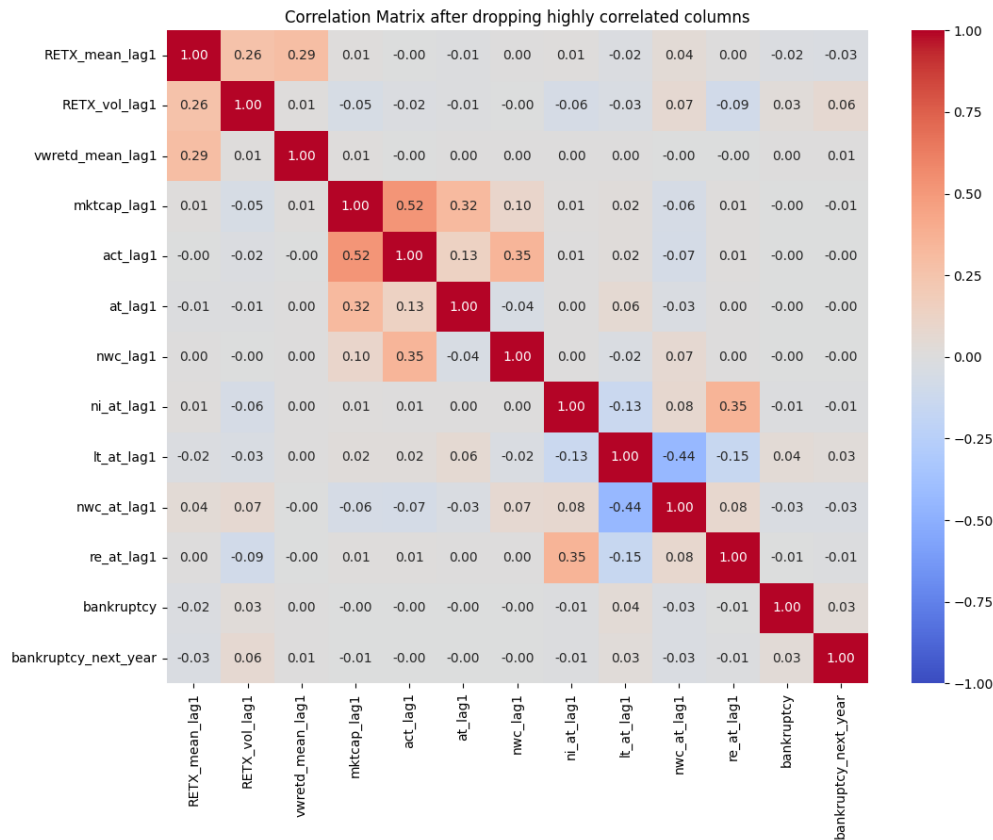


Figure 3: Correlation Matrix after Dropping Highly Correlated Columns

The correlation matrix shows that most explanatory variables are weakly correlated, indicating limited multicollinearity. Market-based features (`RETX_mean_lag1`, `RETX_vol_lag1`, `mktcap_lag1`) are largely independent of accounting-based variables such as `ni_at_lag1`, `lt_at_lag1`, and `nwc_at_lag1`. A moderate negative relationship between leverage (`lt_at_lag1`) and liquidity (`nwc_at_lag1`) reflects typical firm financing trade-offs. No pairwise correlations exceed 0.6, suggesting that all predictors contribute distinct information to the classification models.

1.4 Summary

Overall, the dataset provides:

- A broad time span of firm-level observations, with most data concentrated after 1990.
- Adequate variation in firm characteristics, size, and performance to train and evaluate predictive models effectively.
- Highly unbalanced target label making this a rare event prediction task

- Computationally expensive data
- No strong relation between any predictor and target label

2 Dataset Variables

This section summarizes the key variables derived from Compustat (accounting-based) and CRSP (market-based) data. Each variable is accompanied by its economic interpretation and the expected direction of its relationship with default risk.

2.1 Accounting-Based Ratios (Compustat)

Table 1: Accounting Ratios and Expected Effect on Default Risk

Variable	Definition	Economic Meaning	Effect on Default
ni_at	Net income / Total assets	Profitability relative to firm size; measures efficiency in generating earnings.	—
lt_at	Total liabilities / Total assets	Leverage ratio; proportion of assets financed by debt.	+
nwc_at	Net working capital / Total assets	Liquidity measure; ability to cover short-term liabilities with current assets.	—
re_at	Retained earnings / Total assets	Indicator of accumulated profitability and financial resilience.	—
ebit_at	EBIT / Total assets	Operating performance before interest and taxes; measures core profitability.	—
mkt_lt	Market value / Total liabilities	Market-based solvency ratio; higher values reflect stronger equity market valuation.	—
sale_at	Sales / Total assets	Asset turnover ratio; measures efficiency in generating revenue from assets.	—

2.2 Market-Based Variables (CRSP)

Table 2: CRSP-Derived Market Features and Expected Effect on Default Risk

Variable	Definition	Economic Meaning	Effect on Default
RETX_mean	Average stock return over the year	Reflects firm performance and investor confidence.	–
RETX_vol	Standard deviation of returns	Captures market volatility and perceived firm risk.	+
vwretd_mean	Average market return (benchmark)	Overall market performance; contextual reference.	+
excess_return	Sum of firm return minus market return	Measures relative outperformance vs. market.	–
mktcap	Market capitalization (price $\times$ shares outstanding)	Proxy for firm size and stability.	–

3 Model Performance Summary — Corporate Bankruptcy Prediction

3.1 Dataset Overview

- **Training (in-sample):** 85,773 observations (1964–1990)
- **Testing (out-of-sample):** 314,824 observations (1990–2020)
- **Target:** Bankruptcy indicator ($\sim 0.47\%$ event rate)

3.2 Tree-Based Models

LightGBM

- Out-of-sample accuracy: 0.9953
- Misclassification rate: 0.0047
- ROC-AUC (test): 0.860
- Recall for bankruptcies: 2%

Table 3: LightGBM Classification Metrics

	Precision	Recall	F1-score
Class 0	1.00	1.00	1.00
Class 1	0.04	0.02	0.03

XGBoost

- Out-of-sample accuracy: 0.9967
- Misclassification rate: 0.0033
- ROC-AUC (test): 0.850
- Recall for bankruptcies: 1%

Table 4: XGBoost Classification Metrics

	Precision	Recall	F1-score
Class 0	1.00	1.00	1.00
Class 1	0.12	0.01	0.02

Random Forest

- Best parameters: `max_depth=5, n_estimators=100`
- Cross-validated ROC-AUC (train): 0.8382
- Test ROC-AUC: 0.8753
- Out-of-sample accuracy: 0.9945
- Misclassification rate: 0.0055

Table 5: Random Forest Classification Metrics

	Precision	Recall	F1-score
Class 0	1.00	1.00	1.00
Class 1	0.06	0.03	0.04

3.3 Linear and Penalized Regression Models

Ridge Logistic Regression

- Best λ : 10,000
- ROC-AUC (test): 0.8677
- Out-of-sample accuracy: 0.9824
- Misclassification rate: 0.0176
- Top decile default rate: 2.07%

LASSO + Post-LASSO Logistic

- Selected features: `RETX_mean_lag1, RETX_vol_lag1, ni_at_lag1, lt_at_lag1, nwc_at_lag1`
- ROC-AUC (test): 0.8578
- Best λ : 0.00131
- Out-of-sample accuracy: 0.9811
- Misclassification rate: 0.0189
- Top decile default rate: 1.86%

3.4 KNN

- Best parameters: $k=5$
- Out-of-sample accuracy: 0.9521
- Misclassification rate: 0.0479
- ROC-AUC (test): 0.5448

Table 6: KNN Classification Metrics (Out-of-Sample, $k = 5$)

Metric	Value
Accuracy	0.9521
Misclassification rate	0.0479
Precision (Class 1)	0.011
Recall (Class 1)	0.120
F1-score (Class 1)	0.020
Miss classification rate (label 0)	0.046
Miss classification rate (label 1)	0.880
ROC-AUC (test)	0.5448

3.5 Random Survival Forest (RSF)

Overview

- Purpose: Model time-to-bankruptcy using censored survival framework
- Hyperparameters: `n_trees=500`, `max_depth=8`, `min_samples_leaf=50`
- Features: Post-LASSO variables
- Out-of-sample accuracy: 0.9947
- Misclassification rate: 0.0053
- ROC-AUC (test): 0.879

Table 7: Random Survival Forest Classification Metrics

	Precision	Recall	F1-score
Class 0	1.00	1.00	1.00
Class 1	0.07	0.03	0.04

Table 8: Random Survival Forest Performance

Metric	Value
ROC-AUC (test)	0.879
Cross-validated ROC-AUC (train)	0.841
Top decile default rate	2.41%
Bottom decile default rate	0.05%

Results

3.6 Overall Ranking (Out-of-Sample Discrimination)

Table 9: Model Comparison by Out-of-Sample Discrimination

Model	ROC-AUC
Random Survival Forest	0.879
Random Forest	0.875
Ridge Logistic	0.868
Baseline Logistic	0.861
LightGBM	0.860
Post-LASSO Logistic	0.858
XGBoost	0.850
KNN	0.544

3.7 Key Insights

- All models achieve high accuracy but struggle with rare-event recall.
- Penalized logistic models (Ridge, LASSO) perform competitively with tree-based models.
- Decile analysis confirms monotonicity of predicted probabilities with actual bankruptcy incidence.